

Project Initialization and Planning Phase

Date	5 July 2026
Team ID	
Project Title	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Project Overview	
Objective	To develop a Machine Learning-based Credit Card Approval Prediction System that accurately predicts whether a credit card application should be approved or rejected using applicant information. The system provides a web-based interface, stores prediction history in an SQLite database, displays prediction confidence scores, and offers dashboard analytics to support faster and more reliable decision-making.
Scope	The project analyzes applicant information, performs data preprocessing and feature encoding, trains a Random Forest classification model, and provides real-time credit card approval prediction through a Flask web application. The system also stores applicant prediction history in an SQLite database, displays confidence scores, and provides a dashboard for monitoring prediction statistics and analytics.
Problem Statement	
Description	Manual credit card approval is time-consuming, error-prone, and may result in inconsistent decisions due to the large number of applications received by financial institutions. Managing applicant records manually is difficult and often delays the approval process. An intelligent prediction system is required to automate decision-making while maintaining prediction history and providing useful analytics.
Impact	The proposed system automates the credit card approval process using Machine Learning, reduces manual effort, improves prediction accuracy, stores applicant prediction history in an SQLite database, and provides dashboard analytics for better monitoring and decision-making. This results in faster approvals, improved operational efficiency, and enhanced customer satisfaction.

Proposed Solution	
Approach	The system uses a Random Forest Machine Learning classifier to analyze applicant details such as income, employment status, education, family status, housing type, and financial information to predict whether an applicant is a Good Customer or Bad Customer. A Flask-based web application provides an interactive interface for users to enter applicant details. The prediction result, confidence score, prediction date, and time are automatically stored in an SQLite database. Users can also view previous prediction history and dashboard analytics for monitoring prediction statistics.
Key Features	• Machine Learning-based credit card approval prediction • Data preprocessing and feature engineering • Random Forest classification model • Real-time prediction using Flask • Prediction confidence score • SQLite database integration • Applicant prediction history • Dashboard and analytics • Responsive web interface

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU	Intel i5 or above
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD or above
Software		
Frameworks	Python Web Framework	Flask

Libraries	Machine Learning & Data Analysis	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Pickle
Development Environment	IDE	Jupyter Notebook, VS Code / PyCharm
Database	Database Management	SQLite3
Version Control	Source Code Management	Git & GitHub
Data		
Data	Source, size, format	Credit Card Approval Dataset (CSV), UCI Repository / Kaggle, Applicant Prediction History stored in SQLite Database